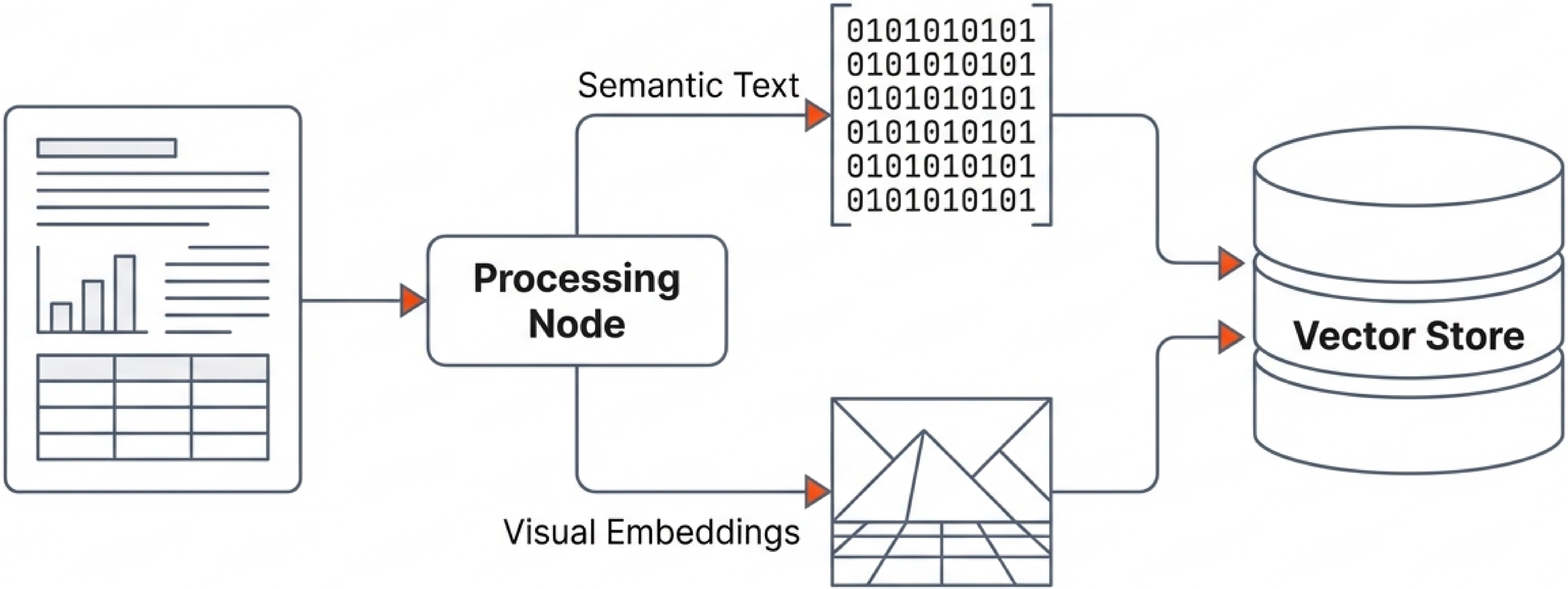
Architecting Next-Gen RAG Systems: Vector Databases & Multimodal Pipelines

A 2025 Engineering Guide to Infrastructure, Extraction Strategies, and Production Implementation



Synthesized from research by Firecrawl, PwC, EPFL (MMORE), and Transurban.

The Evolution of RAG

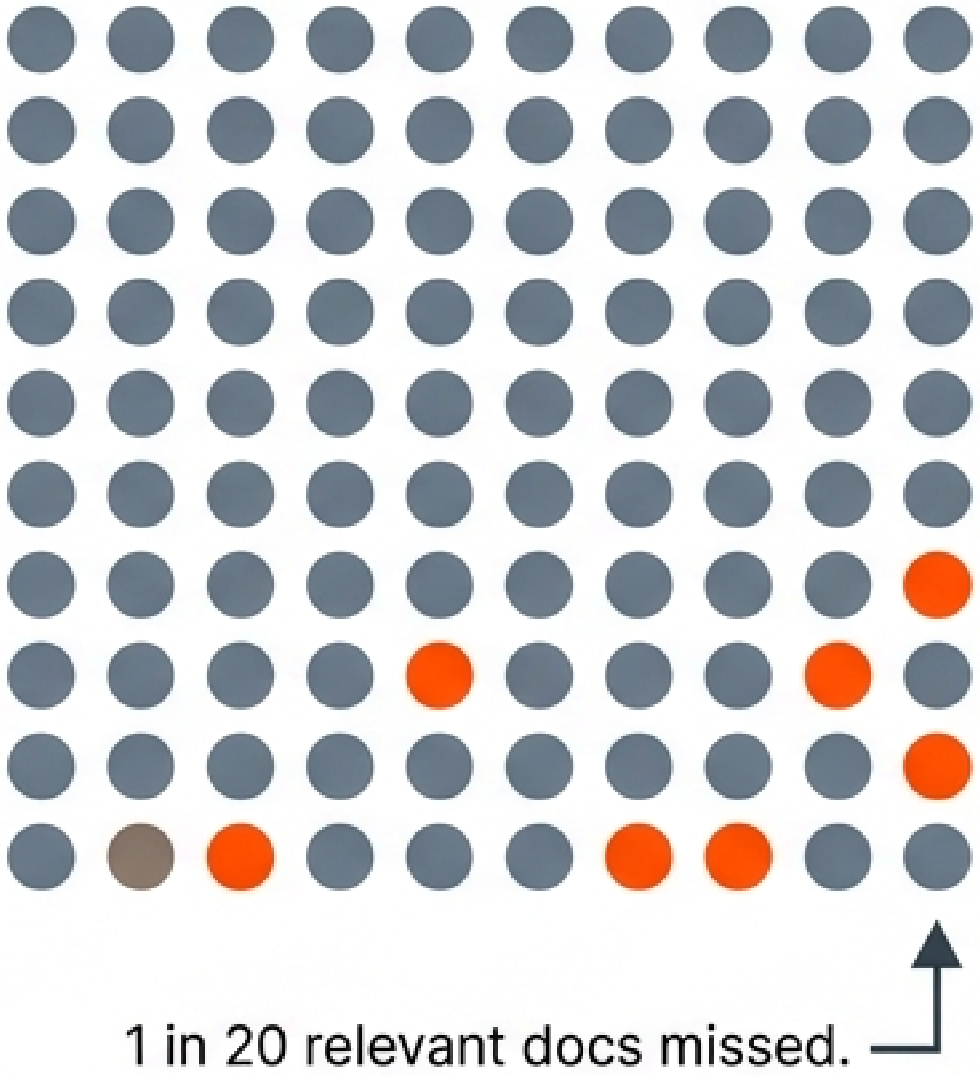
The market has shifted from simple text matching to complex multimodal reasoning.
Current systems have hit a performance ceiling defined by the trade-off between recall and latency.

$10.6B

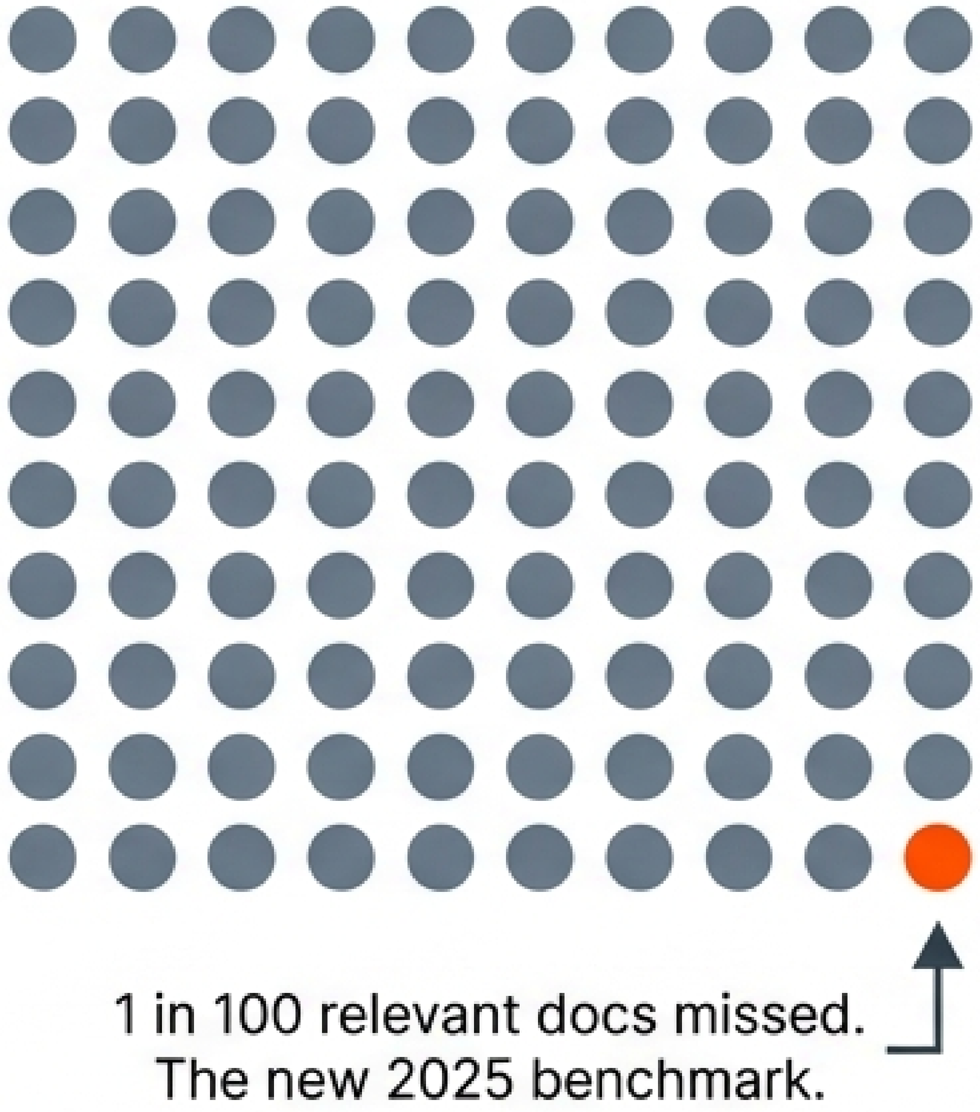
Projected Vector DB Market by 2032

The Recall vs. Speed Trade-off

95% Recall (Gen 1)



99% Recall (Gen 2)



`Target Latency: <50ms`

The 2025 Vector Database Landscape

Pinecone
MANAGED

Best for Time-to-Market.
Serverless auto-scaling.
Zero ops.

Milvus
OPEN SOURCE

Best for Billion-Scale.
Cost-efficient for massive datasets. Self-hosted.

Weaviate
HYBRID

Best for Hybrid Search.
Native keyword + vector filtering.

Qdrant
EDGE / COST

Best Free Tier (1GB).
Compact Rust footprint.
Good for on-device.

pgvector
UNIFIED

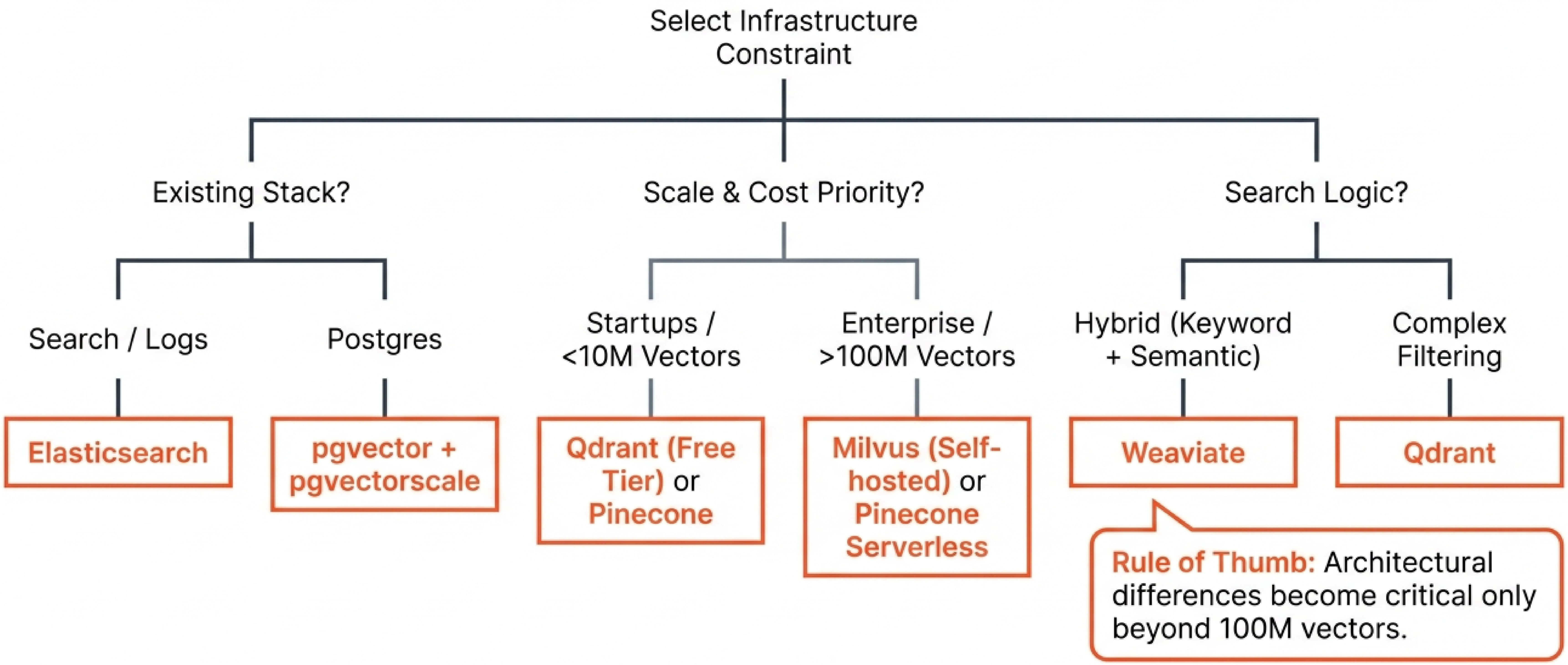
Best for Postgres Shops.
Keep vectors alongside relational data.

Elasticsearch
RELIABILITY

Best for Existing Stacks.
Battle-tested search engine integration.

Engineering Precision

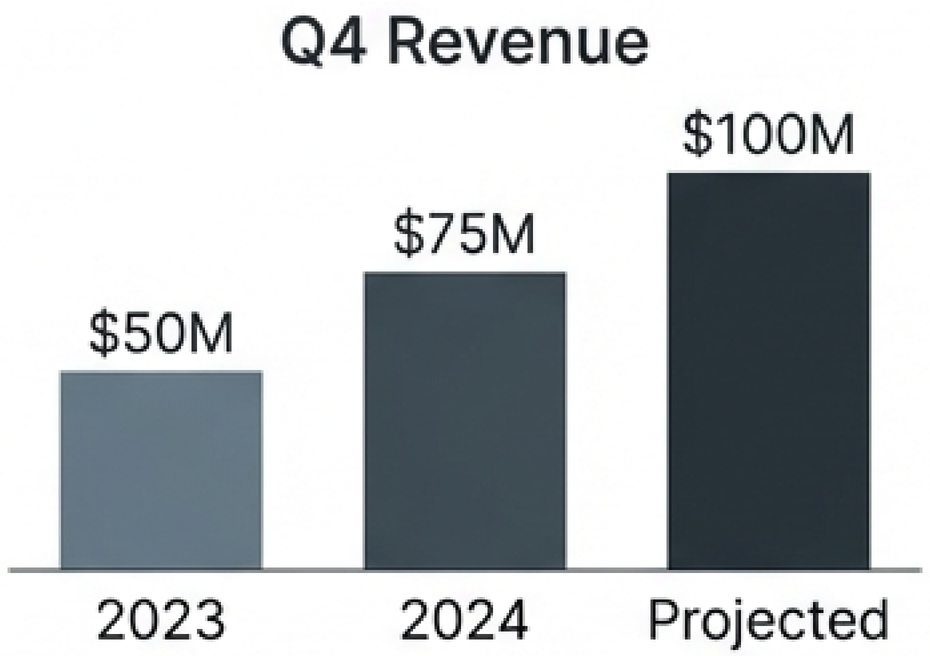
Decision Matrix: Choosing Your Vector Engine



The Multimodal Challenge

Why Text Extraction Fails on Complex Docs

Original Source



Q4	2023	2024	Projected
$M	50	75	100

LLM Text Summary

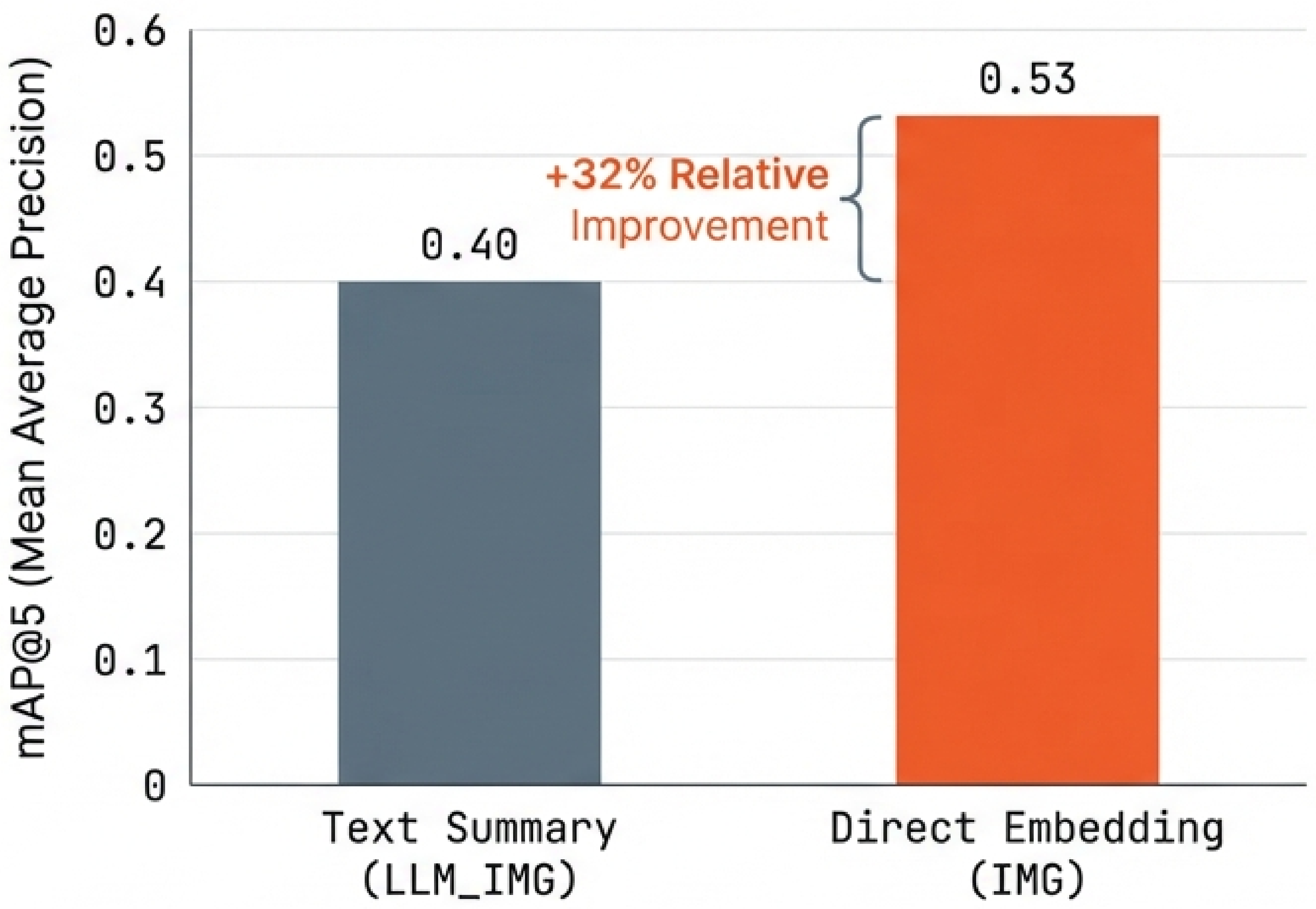
```
The image shows a
chart about revenue.
It goes up over time.
```

"Existing multimodal RAG systems rely on LLM-based summarization... causing loss of contextual information, spatial relationships, and numerical precision." – PwC arXiv

Strategy 1: Direct Multimodal Embedding

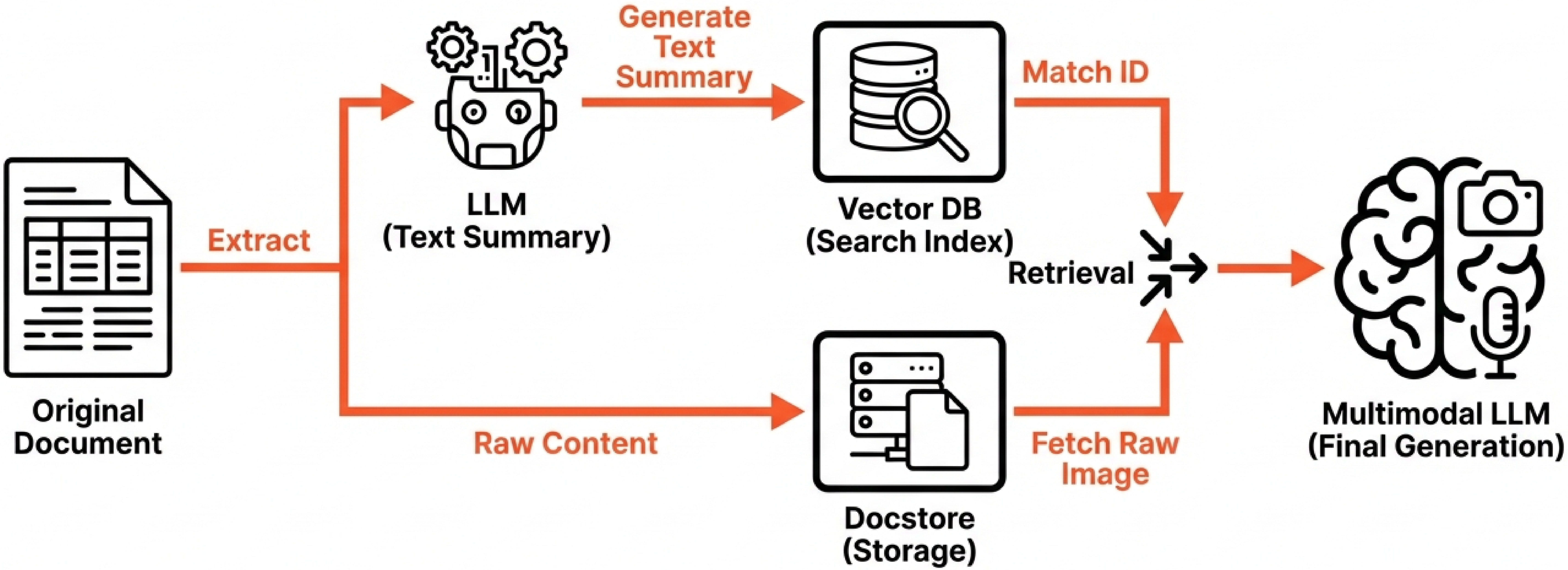
Concept: Embed images and text into a unified vector space (e.g., `Jina v4`, `CLIP`). Images are stored natively.

Best For: High-fidelity visual retrieval.



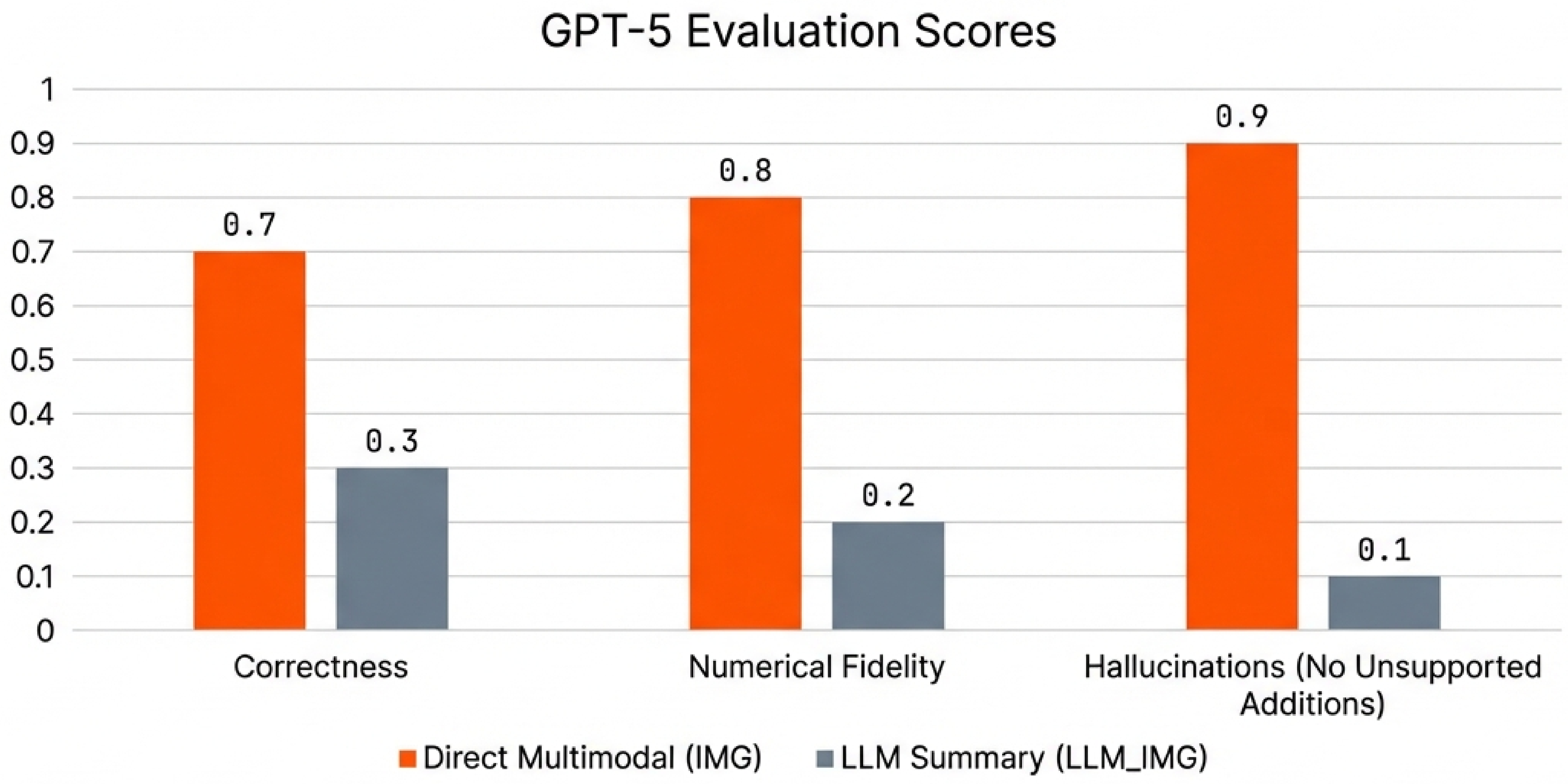
Reduces Hallucinations by ~80% in GPT-5 evaluations.

Strategy 2: The Multi-Vector Retriever (Decoupling)



Decouples the searchable summary from the high-fidelity raw content.

Benchmarking: Direct Embedding vs. Summarization



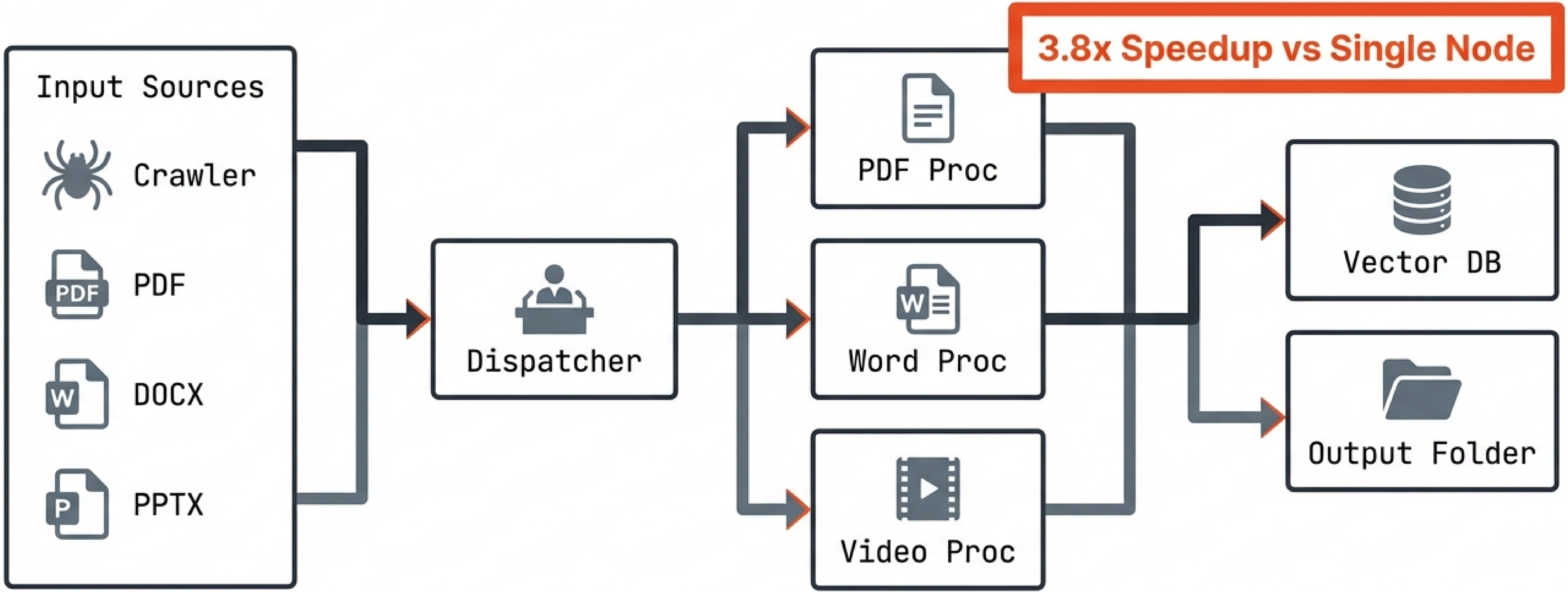
Ingestion Pipeline A: Web & HTML (Firecrawl)

Transforming the messy web into clean vectors.

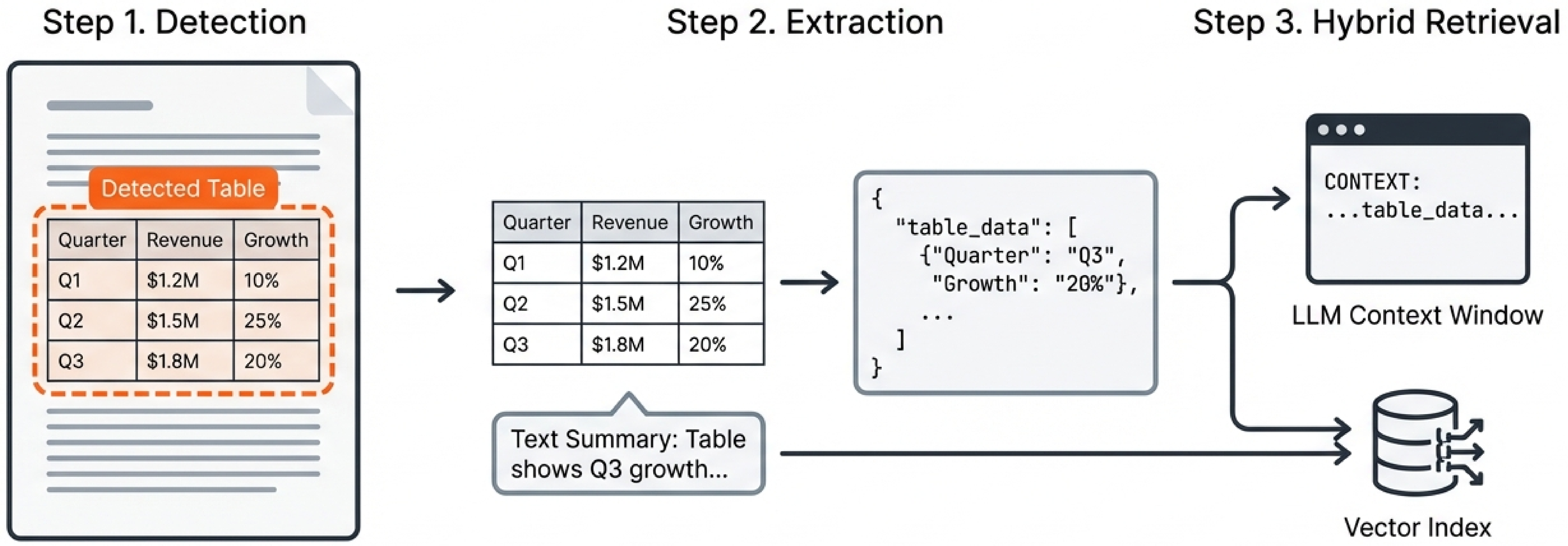

```
app = FirecrawlApp(api_key=key)
data = app.scrape_url(url, params={'formats': ['markdown']})
```

Ingestion Pipeline B: Massive Multimodal (MMORE)

Distributed extraction for heavy enterprise documents.



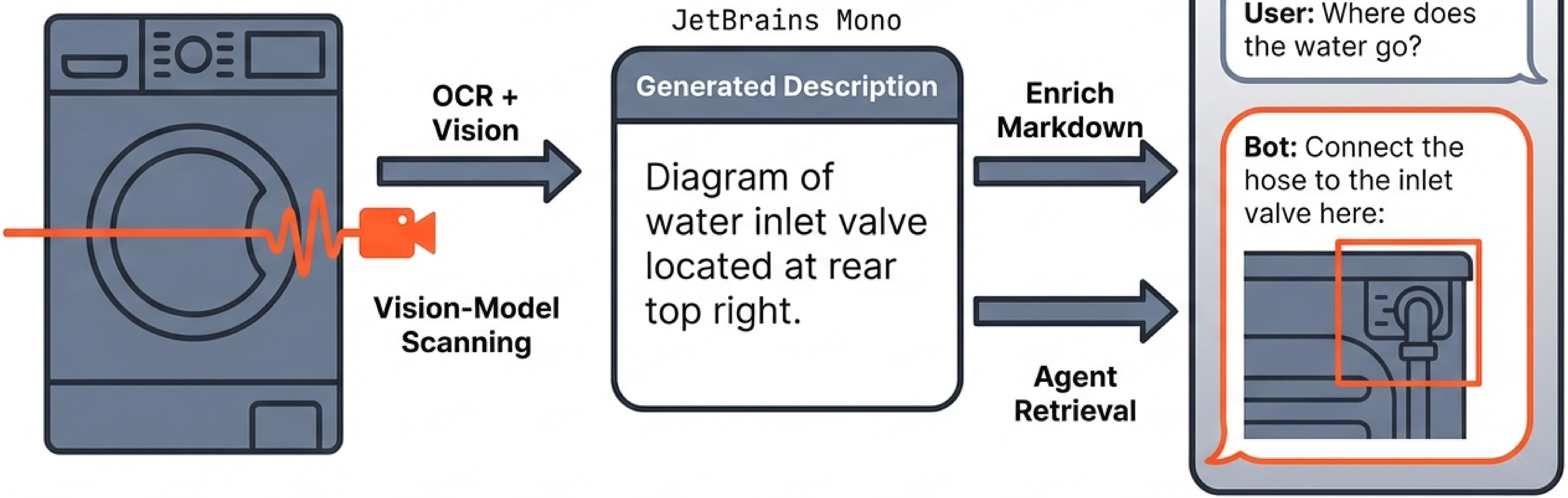
Solving the Table Problem



Naive chunking destroys row/column logic. Extract table structure to **JSON/Markdown** to preserve context.

Building Agents: The n8n & Mistral Approach

Intelligent User Manual



Engineering Precision

Case Study: Transurban's RAGVA

From Rule-Based Chatbot to RAG Assistant

Input Guardrails

PII Masking,
Sanitization

RAG Core

Domain Retrieval,
Context Injection

Output Guardrails

Safety Check,
Format Verification

Engineering Focus:
Last-Mile Testing on domain-specific cases (e.g., Toll Payment).

The RAG Evaluation Framework

Retrieval Metrics

`mAP@5, nDCG@5`

Did we find the right document?

Generation Metrics

`Faithfulness, Hallucination Rate`

Did the LLM answer correctly?

Operational Metrics

`Latency p99, Concurrent QPS`

Is it fast enough for production?

Conclusion & Future Outlook

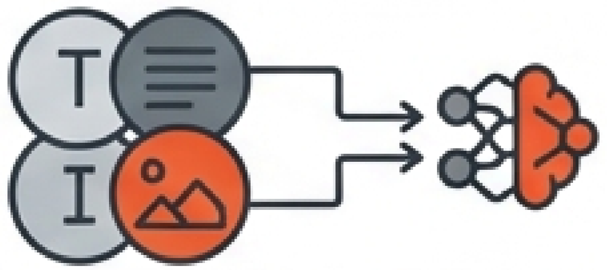
✓ 1. **Infrastructure**
Match Vector DB to scale. Use pgvector for unified stacks, specialized DBs for >100M.

✓ 2. **Modality**
Stop flattening data. Use Direct Embeddings or Multi-Vector decoupling.

✓ 3. **Pipeline**
Garbage-in, Garbage-out. Use robust ingestion (Firecrawl/MMORE).

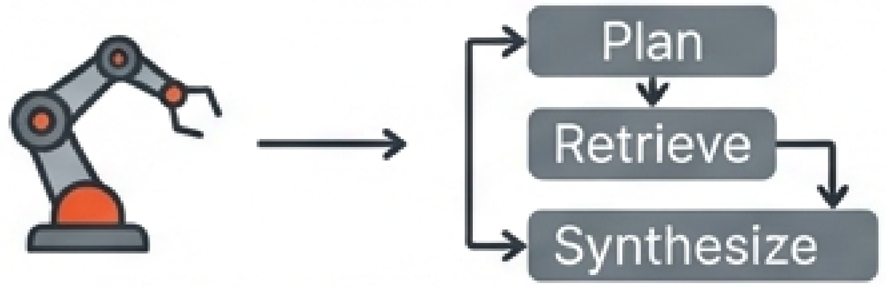
The Future

Unified Models



Universal embeddings (Jina v4) handling text/image natively.

Agentic Planning



Systems that plan retrieval steps rather than just searching once.

Solve for your current scale, but architect for a multimodal future.